Progress Report

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Members

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Current Progress

Since the 04-09 report the project has moved from "RTL simulates, bitstream generates" to **verified hardware on silicon**. On the simulation side, we completed full integration testing for Board A, Board B, and the combined system top, all running with 16 symbols (companies) and resolving timing issues that had affected earlier runs. The most significant milestone is that Board A is now **fully operational on the AUP-ZU3 FPGA** running under PYNQ, with all 16 symbol slots exercised from Python over AXI-Lite MMIO.

RTL Synthesis Fixes

Two classes of synthesis defects were identified and resolved during the Vivado build cycle:

- **Latch elimination (board_a_axi_regs.sv, board_b_axi_regs.sv).** Dynamic integer indexing inside `always_ff` blocks caused Vivado to infer 1,536 latches per register file. Replaced with explicit `for`-loop address matching so every array element is assigned every cycle; combinational read mux moved to a separate `always_comb` block.
- **FDCPE removal (market_sim.sv).** Data-dependent assignments in the asynchronous reset path forced Vivado to emulate 1,536 complex FDCPE cells (UltraScale+ only supports constant async resets). The `init_symbol_tables` task was removed; the asynchronous reset now assigns only constants, and data-dependent initialization was moved to a synchronous path to comply with UltraScale+ flip-flop requirements (`market_sim.sv`, `lfsr32.sv`).

After these fixes both Board A and Board B synthesise and implement with **zero critical warnings**, zero latches, and timing met at 50 MHz PL clock.

PYNQ Deployment Flow

We transitioned to a **full PYNQ overlay workflow**. A persistent blocker was a faulty Vivado block design that prevented AXI reads and writes from reaching our custom IP. After extensive debugging, we resolved this by introducing a `board_a_top_bd.v` wrapper and restructuring the block design's clock, reset, and AXI interconnect wiring in Vivado. With a working PS–PL path, we were able to deploy `.bit`/`.hwh` overlay pairs to each board's SD card and interact with the hardware from PYNQ's Jupyter environment. We also updated Python scripts (`config_symbols.py`, `symbol_config_panel.py`, `telemetry_server.py`) to use correct IP block names and dynamic MMIO address ranges from the overlay's `ip_dict`.

Board A Hardware Verification

Board A has been tested on silicon using Python scripts executed from PYNQ's Jupyter environment over AXI-Lite MMIO. Key results:

- **Register read/write:** All scalar registers and all 16 per-symbol array registers (init_mid, init_spread, sector_id) write and read back correctly.
- **Market simulator:** Configured 4 and 16 symbols, started the simulator via CTRL, and confirmed ∼68,300 quotes/s at 50 MHz with `QUOTE_INTERVAL`=1000. Quote counter increments steadily across multiple seconds.
- **Regime switching:** Cycled through all four regimes (CALM, VOLATILE, BURST, ADVERSARIAL) while running; RGB0 LED color changes confirmed visually on the board.
- **FSM reset/restart:** CTRL[1] reset pulse stops the simulator and clears counters to zero; CTRL[0] restart brings it back online.
- **Board B:** We have successfully generated `.bit` and `.hwh` files for Board B; however, hardware testing has not yet been performed.

Remaining Work

B1. Run Board B standalone PS tests: AXI register readback, strategy selection, position/cash registers, histogram readout.

B3. Verify Board B LEDs and RGB indicators for FSM states (IDLE/ARMED/TRADING/HALTED).

S1. Connect Board A and Board B via PMOD cable; verify `link_up` on both boards.

S2. Confirm quote flow: Board A `quotes_sent` vs

Board B `quotes_rcvd`.

S3. Enable trading on Board B; verify orders sent, fills received, and position updates.

S4. Run `telemetry_server.py` on Board B and connect laptop dashboard for live visualization.

S5. Run multi-regime demo: sweep CALM → VOLATILE → BURST → ADVERSARIAL while both boards are connected, observe strategy adaptation and P&L behavior.

D1. Prepare final demo script and presentation materials.

Individual Contributions

Kushal Agrawal

Led **integration testing** for Board A, Board B, and the combined system top with 16 symbols, resolving timing issues from earlier runs. Diagnosed and resolved synthesis defects (latch elimination in AXI register files, FDCPE removal in `market_sim.sv`) with RTL refactoring. Updated Python deployment scripts for correct IP block naming and MMIO address handling.

Outstanding

Deploy Board B overlay and run standalone hardware tests; connect both boards via PMOD and complete system-level integration (S1–S5).

Eeshana Hari

Led **PYNQ deployment and hardware bring-up** for Board A: flashed SD cards, configured USB gadget connectivity, uploaded overlays and Python scripts to Jupyter Lab, and executed all on-hardware tests. Debugged and restructured the Vivado block design to establish a working PS–PL AXI path. Continued ownership of **Board A RTL** and **link layer** modules.

Outstanding

Lead Board B hardware bring-up; coordinate inter-board PMOD link testing and system-level demo preparation.